

Session 8: Introduction to PL/SQL Trainer Coverage: PL/SQL block structure (DECLARE, BEGIN, EXCEPTION, END). Variables and cursors. Control structures (IF, LOOP). Demo: Simple stored procedure to calculate user's total spend.

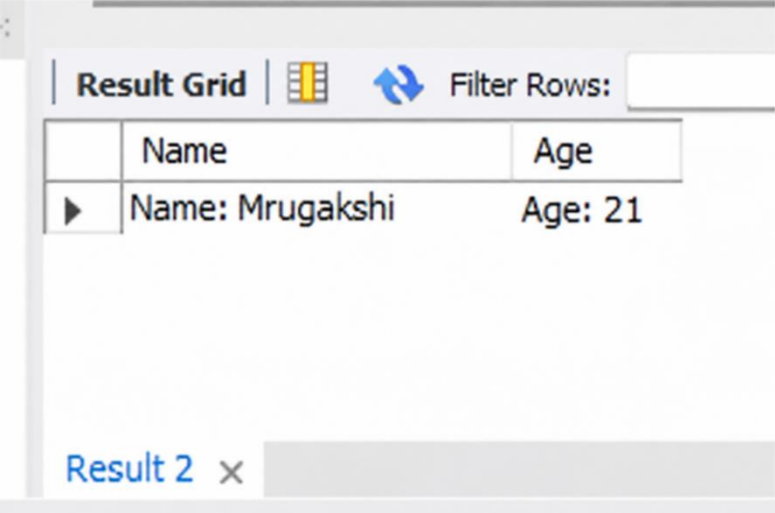
Question1: Write a PL/SQL block that declares two variables: one for a user's name (VARCHAR2) and one for their age (NUMBER). Assign values and print both using DBMS_OUTPUT.PUT_LINE.

Ans:

```
SET @user_name = 'Mrugakshi';
```

```
SET @user_age = 21;
```

```
SELECT CONCAT('Name: ', @user_name) AS Name, CONCAT('Age: ', @user_age)  
AS Age;
```



The screenshot shows a 'Result Grid' window with a table containing two columns: 'Name' and 'Age'. The first row of data shows 'Name: Mrugakshi' and 'Age: 21'. The window has a 'Filter Rows' button and a tab labeled 'Result 2'.

	Name	Age
▶	Name: Mrugakshi	Age: 21

Question2: Create a PL/SQL block that uses an IF statement to check if a given order amount is above 500. If yes, print 'Eligible for free delivery', else print 'Delivery charges apply'.

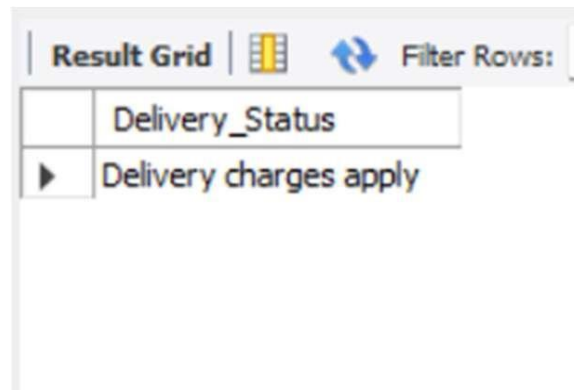
Ans:

```
set @order_amount=500;
```

```
select if(@order_amount >
```

```
500,
```

'Eligible for free
delivery', 'Delivery
charges apply') as
Delivery_Status;



The image shows a screenshot of a database query result grid. The grid has two columns and two rows. The first row has a small square icon in the first column and the text 'Delivery_Status' in the second column. The second row has a small square icon with a right-pointing arrow in the first column and the text 'Delivery charges apply' in the second column. Above the grid, there is a header bar with the text 'Result Grid' and a 'Filter Rows:' button.

	Delivery_Status
	Delivery charges apply

Question3: Write a PL/SQL block that uses a simple LOOP to print the numbers 1 to 5 using

**DBMS_OUTPUT.PUT_LINE.

Hint:**

Use a counter variable and EXIT WHEN condition inside the loop. Ans:

USE counter_mydb;

DELIMITER //

CREATE PROCEDURE print_numbers()

BEGIN

DECLARE counter INT DEFAULT 1;

simple_loop: LOOP

SELECT counter;

SET counter = counter + 1;

IF counter > 5 THEN

LEAVE simple_loop;

```

        END IF;

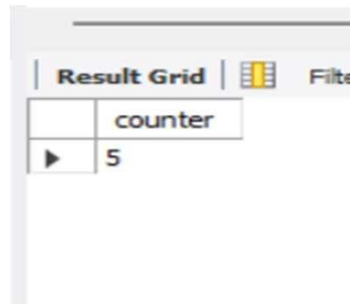
    END LOOP;

END //

DELIMITER ;

CALL print_numbers .

```



	counter
▶	5

Question4: Declare a cursor in PL/SQL to select all product names from a table called PRODUCTS. Fetch each product name and print it using DBMS_OUTPUT.PUT_LINE.

Hint: Assume the PRODUCTS table has a column PRODUCT_NAME.

Ans:

```
USE USE mydb;
```

```

DROP PROCEDURE IF EXISTS
show_product_names;

```

```

INSERT INTO PRODUCTS (PRODUCT_NAME)
VALUES

```

```
('Laptop'),
```

```
('Mobile'),
```

```
('Keyboard'),
```

```
('Mouse'),
```

```
('Headphones');
```

```
DELIMITER //

CREATE PROCEDURE show_product_names()

BEGIN

    DECLARE done INT DEFAULT 0;

    DECLARE productName VARCHAR(255);


    DECLARE product_cursor CURSOR FOR

        SELECT PRODUCT_NAME FROM
PRODUCTS

        WHERE PRODUCT_NAME = 'Laptop';


    DECLARE CONTINUE HANDLER FOR NOT
FOUND SET done = 1;


    OPEN product_cursor;


    read_loop: LOOP

        FETCH product_cursor INTO productName;


        IF done = 1 THEN

            LEAVE read_loop;

        END IF;


        SELECT productName;

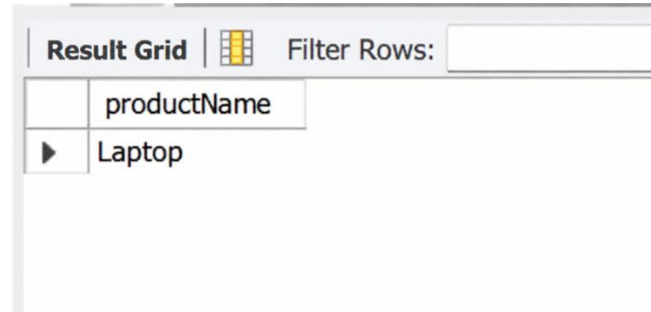
    END LOOP;


    CLOSE product_cursor;

END //
```

DELIMITER ;

CALL show_product_names();



	productName
▶	Laptop

Question5: Write a PL/SQL block that tries to divide two numbers entered as variables. If a division by zero occurs, catch the exception and print 'Cannot divide by zero'. Otherwise, print the result.

Ans:

USE mydb;

DROP PROCEDURE IF EXISTS divide_numbers;

DELIMITER //

CREATE PROCEDURE divide_numbers()

BEGIN

DECLARE num1 DECIMAL(10,2) DEFAULT 10;

DECLARE num2 DECIMAL(10,2) DEFAULT 0;

DECLARE result DECIMAL(10,2);

DECLARE CONTINUE HANDLER FOR SQLEXCEPTION

BEGIN

SELECT 'Cannot divide by zero' AS Message;

END;

IF num2 = 0 THEN

SIGNAL SQLSTATE '45000'

SET MESSAGE_TEXT = 'Cannot divide by zero';

ELSE

```
SET result = num1 / num2;
```

```
SELECT result AS Result;
```

```
END IF;
```

```
END //
```

```
DELIMITER ;
```

```
CALL divide_numbers();
```

Result Grid		Filter Rows:
	Message	
▶	Cannot divide by zero	